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1. TITLE OF THE PROJECT

“ONLINE INTERNSHIP & FREELANCE PORTAL”

2. INTRODUCTION AND OBJECTIVES OF THE PROJECT

2.1 INTRODUCTION

The **Online Internship & Freelance Portal** is a comprehensive web-based platform designed to connect students with companies offering internships and freelance opportunities. The system helps bridge the gap between skilled students and industry needs by providing a seamless platform for applications, tests, assignments, payments, messaging, and performance evaluation.

The portal supports three main user roles — **Admin, Company, and Student**. Students can create profiles, upload resumes, manage personal skills, browse opportunities, apply for internships or freelance work, submit tests and assignments, and communicate with companies. Companies can create opportunities, define required skills, evaluate applicants, conduct tests, assign project work, and track submissions. Admins can monitor system activity, manage users, approve company registrations, handle payments, view reports, and maintain platform integrity.

The system integrates user authentication, skills mapping, opportunity posting, workflows for applications and evaluations, real-time messaging, notifications, and payment tracking. It also includes feedback and review capabilities to ensure transparency and quality of service. Overall, the platform provides a secure, efficient, and user-friendly experience for both students seeking opportunities and companies seeking talent.

2.2 OBJECTIVES

- ❑ **Centralized User Management** – To provide a unified system for handling students, companies, and administrators with proper authentication and authorization.
- ❑ **Opportunity Management** – Allow companies to create, update, and manage internship and freelance opportunities with required skills and duration.
- ❑ **Skill Mapping** – Ensure accurate mapping between student skills and company-required skills to improve the quality of applications.
- ❑ **Application & Selection Workflow** – Provide a structured process for applications, shortlisting, hiring, and tracking student status.
- ❑ **Test Management** – Allow companies to upload tests, set deadlines, and evaluate submissions with marks and remarks.
- ❑ **Assignment Tracking** – Enable companies to assign project work to students, track submission progress, and grade assignments.
- ❑ **Communication System** – Provide real-time messaging between students and companies for clarification and updates.
- ❑ **Payment Management** – Maintain transparency in payments between students and companies for various opportunities.
- ❑ **Feedback & Review System** – Allow students to review companies and submit feedback to the admin, improving service quality.

- **Notifications** – Real-time alerts for applications, test deadlines, assignment updates, messages, and payments.
- **Data Security & Privacy** – Ensure secure user authentication, data protection, and controlled access.
- **Scalability** – Design the system to support a growing number of users, companies, and opportunities.

3. PROJECT CATEGORY

This project belongs to the category of **Web-based Application using RDBMS (Relational Database Management System)**.

An RDBMS stores data in structured tables with rows and columns, enabling relationships between entities such as users, skills, opportunities, applications, tests, and payments. Relational databases allow efficient querying, data redundancy reduction, and strong integrity through primary and foreign key constraints.

4. TOOLS/PLATFORM SOFTWARE AND HARDWARE SPECIFICATIONS

4.1 TOOLS/PLATFORM

FRONT END: HTML, CSS, JAVASCRIPT

Used for the user interface, form validations, responsive design, and dynamic interactions.

BACK END: PYTHON

Python provides a robust, flexible, and readable backend development environment.

DATABASE: MySQL

A widely used relational database that supports efficient query execution and data integrity.

FRAMEWORK: DJANGO

A high-level Python framework used to build secure, scalable web applications with built-in admin panel, ORM, and authentication.

4.2 SOFTWARE SPECIFICATION

Component	Specification
Operating System	Windows 10
Front End	HTML, CSS, JavaScript
Back End	Python
Framework	Django
Database	MySQL
Browsers	Chrome, Firefox, Edge
IDE	PyCharm/VS Code

4.3 HARDWARE SPECIFICATION

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB minimum
Hard Disk	1 TB
Monitor	LED
Keyboard	Standard
Mouse	Optical

5. PROBLEM DEFINITION, REQUIREMENT SPECIFICATION

5.1 PROBLEM DEFINITION

5.1.1 EXISTING SYSTEM

In the traditional internship and freelance hiring process, most tasks are handled manually. Students search for opportunities through multiple sources such as job boards, college notice boards, or social media. Companies often manage applications through emails, spreadsheets, or physical documents, making it difficult to track applicants efficiently.

Communication between students and companies is usually scattered across emails and messaging platforms, resulting in delays and misunderstandings. Tests and assignments are often sent manually through emails without proper deadline tracking or submission records.

Payments, if involved, lack transparency and proper documentation. There is no centralized system where students can track applications or where companies can evaluate applicants in a structured manner. Administrators lack tools for monitoring activities, verifying companies, or generating reports.

This fragmented approach leads to inefficiency, errors, miscommunication, and delays in the hiring process.

5.1.2 PROPOSED SYSTEM

The **Online Internship & Freelance Portal** resolves these challenges by providing a unified, centralized, web-based platform. Students and companies can register, create profiles, and manage activities through a modern interface.

Companies can post opportunities, specify required skills, schedule tests, evaluate submissions, assign projects, and communicate directly with applicants. Students can apply for opportunities, upload resumes, submit tests and assignments, track application status, chat with companies, and manage their skills.

Admins can oversee the entire system, activate/deactivate users, approve companies, monitor payments, view analytics, and manage notifications.

The system reduces manual workload, eliminates errors, and improves transparency by automating key tasks such as application tracking, evaluation, messaging, notifications, and payment monitoring.

By integrating all features into a single platform, it enhances user experience and speeds up the internship/freelance hiring process.

5.2 SOFTWARE REQUIREMENT SPECIFICATION

1. Introduction

1.1 Purpose

The purpose of this project is to develop a comprehensive, web-based **Online Internship & Freelance Portal** that connects students and companies on a centralized platform. The system enables students to create profiles, upload resumes, manage skills, apply for opportunities, take tests, complete assignments, and communicate with companies. Companies can post internships or freelance projects, evaluate candidates through tests, assign work, monitor submissions, make payments, and review student performance. Admins oversee all users, opportunities, payments, applications, reviews, and system-wide activities.

The system aims to eliminate manual recruitment processes and provide an efficient, scalable, secure, and user-friendly digital platform.

1.2 Scope

The system will provide the following features:

- **User Management** (Admins, Students, Companies)
- **Skill Management** for both students and companies
- **Posting Internships & Freelance Opportunities**
- **Application Management** with tracking
- **Online Test Management** (upload tests, submit answers, evaluate)
- **Assignment Handling** (assign tasks, upload submissions, approve work)
- **Integrated Messaging** between students and companies
- **Payment Management** for project-related transactions
- **Reviews & Ratings**
- **Notifications System**
- **Feedback System**
- Dashboard, Reports, and monitoring for Admins

The entire system will be accessible via web browsers and optimized for all devices.

1.3 Definitions, Acronyms, and Abbreviations

- **UI** – User Interface
- **DBMS** – Database Management System
- **SRS** – Software Requirements Specification
- **Admin** – System Administrator
- **ATS** – Applicant Tracking System
- **CRUD** – Create, Read, Update, Delete

1.4 References

- Standard internship portals like Internshala, Naukri, LinkedIn
- IEEE SRS Standard (IEEE 830-1998)

2. Overall Description

2.1 Product Perspective

This Online Internship & Freelance Portal acts as a unified digital solution replacing scattered, manual or semi-digital hiring processes. Students, companies, and administrators interact through a centralized system designed to simplify applications, skill matching, testing, assignment handling, payments, and communication.

2.2 Product Functions

Major functions include:

- **User Registration & Authentication** (Student/Company/Admin)
- **Profile & Skill Management**
- **Posting Opportunities** (Internship/Freelance)
- **Required Skill Mapping**
- **Student Application Tracking**
- **Test Upload, Test Submission & Evaluation**
- **Assignment Creation, Submission & Feedback**
- **Chat/Messaging Module**
- **Payment Processing**
- **Rating & Reviews**
- **Notifications**
- **Admin Dashboard & Reports**

2.3 User Classes and Characteristics

Admin

- Full system control
- Manage users, opportunities, payments, skills
- Generate reports and analytics

Company

- Manage company profile
- Post internships/freelance projects
- Shortlist applicants
- Upload tests, evaluate submissions
- Assign project work
- Chat with students
- Rate students

Student

- Manage personal profile
- Upload resume
- Add skills
- Apply for internships/freelance jobs
- Give tests, upload assignments
- Chat with companies
- Make/view payments
- Give company reviews

2.4 Operating Environment

- **Frontend:** HTML, CSS, JavaScript (Bootstrap optional)
- **Backend:** Python Django Framework
- **Database:** MySQL
- **Server:** Apache/Nginx
- **Platform:** Web-based, mobile responsive

2.5 Constraints

- Should work across modern browsers (Chrome, Firefox, Edge, Safari)
- Encrypted passwords and secure login sessions
- Scalable to handle multiple user roles simultaneously
- Valid file formats for resumes, tests, submissions

2.6 Assumptions and Dependencies

- Users have internet access
- Payment integration needs third-party service
- Email/SMS notifications require external APIs

3. Specific Requirements

3.1 Functional Requirements

1. User Registration & Login

- Students/Companies/Admins can create accounts
- Forgot password functionality

2. Student Module

- Manage profile, resume, skills
- Browse opportunities
- Apply for internships/freelancing
- View application status
- Download tests, upload submissions
- Upload assignment submissions
- Chat with companies

- Make/view payments
- Write reviews

3. Company Module

- Create/edit profile
- Add company skills
- Post opportunities
- Add required skills
- View student applications
- Upload tests
- Evaluate test submissions (marks/remarks)
- Create assignments
- Track assignment submissions
- Approve/reject work
- Chat with students
- View payment history
- Receive reviews

4. Admin Module

- Activate/deactivate users
- Approve company registrations
- View student skills
- Track applications & tests
- Monitor opportunities (delete inappropriate ones)
- Manage payments
- Manage notifications
- View feedback
- Reporting Dashboard:
 - Total students
 - Total companies
 - Total opportunities
 - Applications & hiring stats
 - Payment summaries

5. Test Management

- Upload test files
- Set deadlines
- Students submit answers
- Companies evaluate and give marks

6. Assignment Management

- Create assignment for selected students
- Students upload submission
- Companies mark as completed

7. Messaging System

- Chat between student and company

8. Payments

- Students or companies can make/pay invoices
- Payment status monitoring
- Generate receipts

9. Reviews

- Students give ratings to companies
- Companies review performance of students

10. Notifications

- New applications
- Test deadlines
- Assignment updates
- Payment alerts
- New messages

11. Feedback

- Students and companies can send feedback to admin

3.2 Non-Functional Requirements

Performance

- Must support **200–500 concurrent users**
- Quick loading time (< 3 seconds average)

Security

- Encrypted passwords
- Secured via HTTPS
- SQL Injection, CSRF protection
- Role-based access control

Usability

- Responsive UI
- Easy navigation
- Clear dashboard for all users

Scalability

- Support future expansions (more modules, mobile app)

Maintainability

- Modular code structure (Django MVC)

4. Conclusion

The **Online Internship & Freelance Portal** offers a complete digital transformation of the internship and freelance hiring process. The system centralizes student, company, and admin operations with a powerful set of tools including applications tracking, skill matching, test evaluation, assignment management, payments, reviews, and messaging.

Companies benefit from automated hiring workflows, while students gain seamless access to opportunities. Admins maintain full oversight through monitoring tools, payment tracking, and reporting dashboards.

Overall, the system ensures:

- Improved transparency
- Faster hiring cycles
- Enhanced communication
- Reduced administrative workload
- A scalable platform suitable for academic institutions, companies, and freelancing ecosystems

5.3 PROJECT PLANNING AND SCHEDULING

5.3.1 GANTT CHART

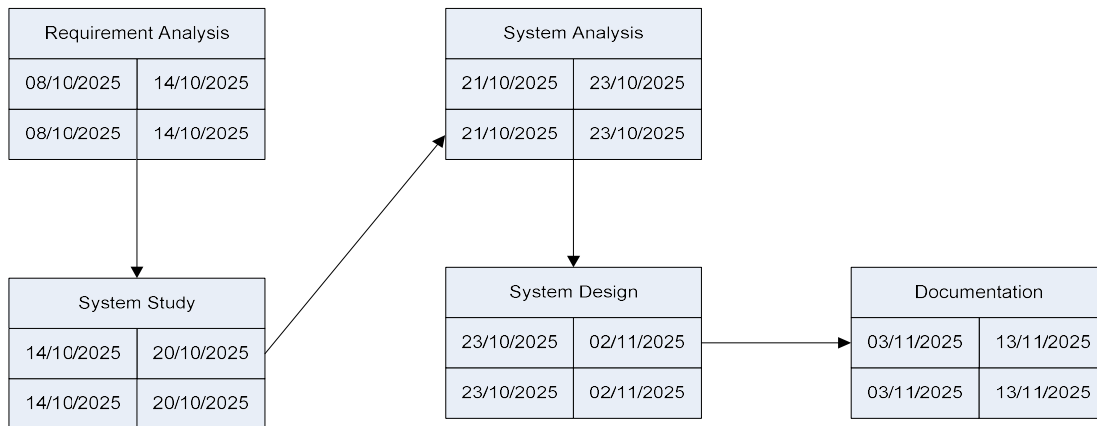
A Gantt chart is a type of bar chart that is used to visually display the schedule of a project. It is used in project management activities to display the project activities, task and events against time. A Gantt chart makes it simple to create, view and monitor project activities and tasks over a given time. Typically, each activity, process or task in a Gantt chart is represented by a horizontal bar scaled parallel to calendar and/or dates. The start and end of each bar represent the start and the end of that particular activity.

ID	Task Name	Start	Finish	Duration	Oct 2025			Nov 2025		
					10-12	10-19	10-26	11-2	11-9	
1	Requirement Analysis	08-10-2025	14-10-2025	1w						
2	System Study	14-10-2025	20-10-2025	1w						
3	System Analysis	21-10-2025	23-10-2025	.43w						
4	System Design	23-10-2025	02-11-2025	1.5w						
5	Documentation	03-11-2025	13-11-2025	1.57w						

5.3.2 PERT CHART

PERT chart is a project management tool used to schedule, organize and coordinate task within a project. Pert stands for Program Evaluation Review Technique. Pert chart presents a graphic illustration of a project as a network diagram consisting of numbered nodes (either circles or

rectangles) representing events, or milestones in the project linked by labelled vectors (directional lines) representing task in the project. The direction of the arrows on the lines indicates the sequence of tasks.



6. SCOPE OF THE SOLUTION

The scope of the *Online Internship & Freelance Portal* is to provide a comprehensive digital platform that connects students, job seekers, interns, freelancers, and employers on a single system. The solution aims to simplify the process of finding internships, freelance projects, and skilled talent while improving the efficiency of recruitment and project management.

The system will support secure user registration and role-based access for administrators, companies, freelancers, and students. It will allow organizations to post internship openings, freelance projects, job opportunities, and skill-based tasks. Users will be able to browse listings, apply for positions, submit proposals, upload resumes or portfolios, and communicate with employers in real time.

The platform will include features for scheduling interviews, shortlisting candidates, assigning freelance tasks, tracking project progress, and maintaining professional profiles. Integrated payment modules will handle freelance work payments, project milestones, invoices, receipts, and transaction history for secure and transparent financial processes.

Additionally, the portal will function as an engagement hub by publishing blogs, success stories, announcements, and notifications. It will also maintain a media gallery to highlight events, workshops, or achievements. From the administrative perspective, the system will manage user verifications, job posting approvals, skill assessments, and platform analytics.

7. ANALYSIS

System analysis is the study of a set of interacting entities within a system, including software components and user workflows. It is closely related to requirement analysis and operations research. The process involves collecting real-world data, understanding existing challenges, identifying user needs, and recommending improvements to enhance system efficiency.

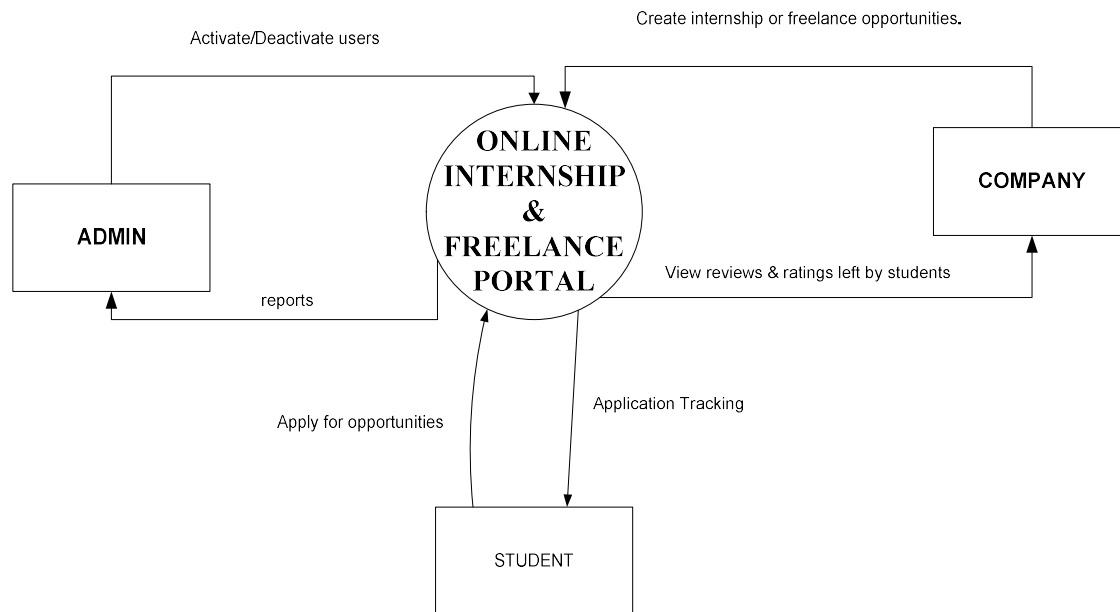
For the *Online Internship & Freelance Portal*, system analysis includes understanding how students, freelancers, and employers interact with the platform, how data flows among

modules like internship posting, applications, task assignments, messaging, payments, and reports, and how current processes can be optimized through automation and digital management.

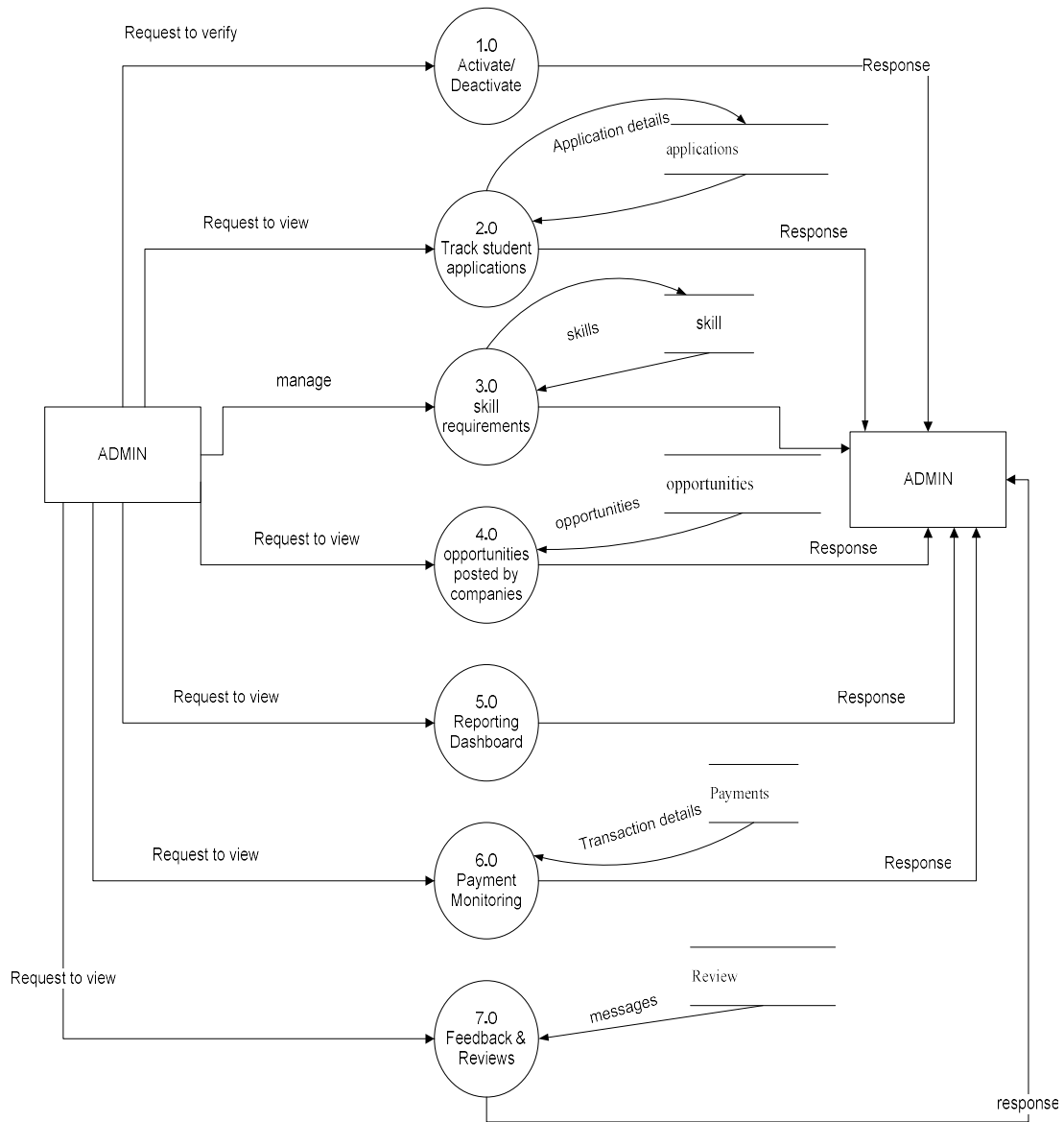
7.1 DATAFLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual tool used to represent the flow of information within a system. In the context of the Online Internship & Freelance Portal, the DFD helps illustrate how data moves between users, processes, and data stores, making it easier to understand the functional structure of the system. It shows how external entities such as students, freelancers, and employers interact with the platform by submitting applications, posting internships, uploading tasks, or making payments. The internal processes, such as user authentication, job posting management, application processing, communication handling, and payment verification, transform and route the data to the appropriate data stores, including user profiles, job listings, applications, messages, and transaction records. By representing the system in terms of data movement rather than detailed program logic, the DFD provides a clear and simplified view of the portal's functionality. It is especially useful during preliminary system design, as it allows analysts and stakeholders to visualize processes, identify missing requirements, and ensure that data flows smoothly throughout the platform. Overall, the DFD serves as an essential documentation component that bridges the communication gap between developers, analysts, and users while supporting an accurate and efficient system design.

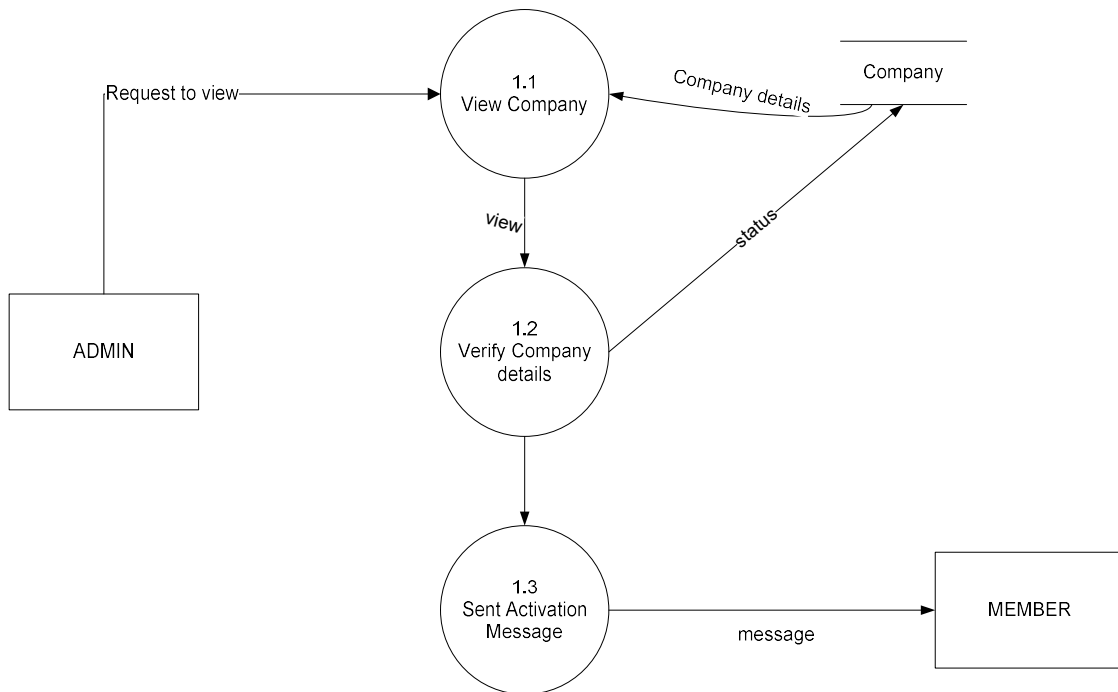
CONTEXT LEVEL DIAGRAM



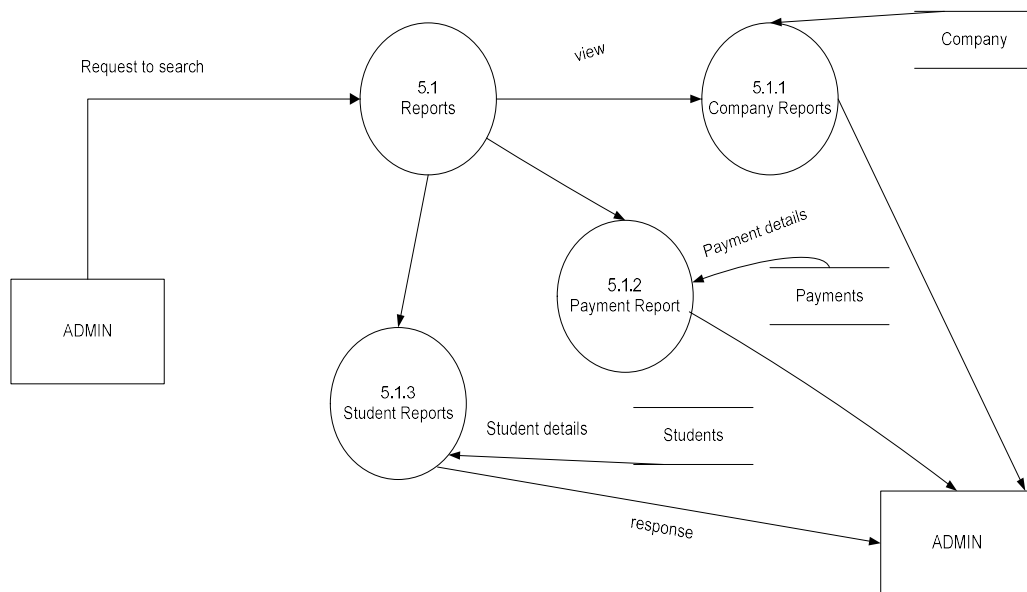
FIRST LEVEL DFD FOR ADMIN



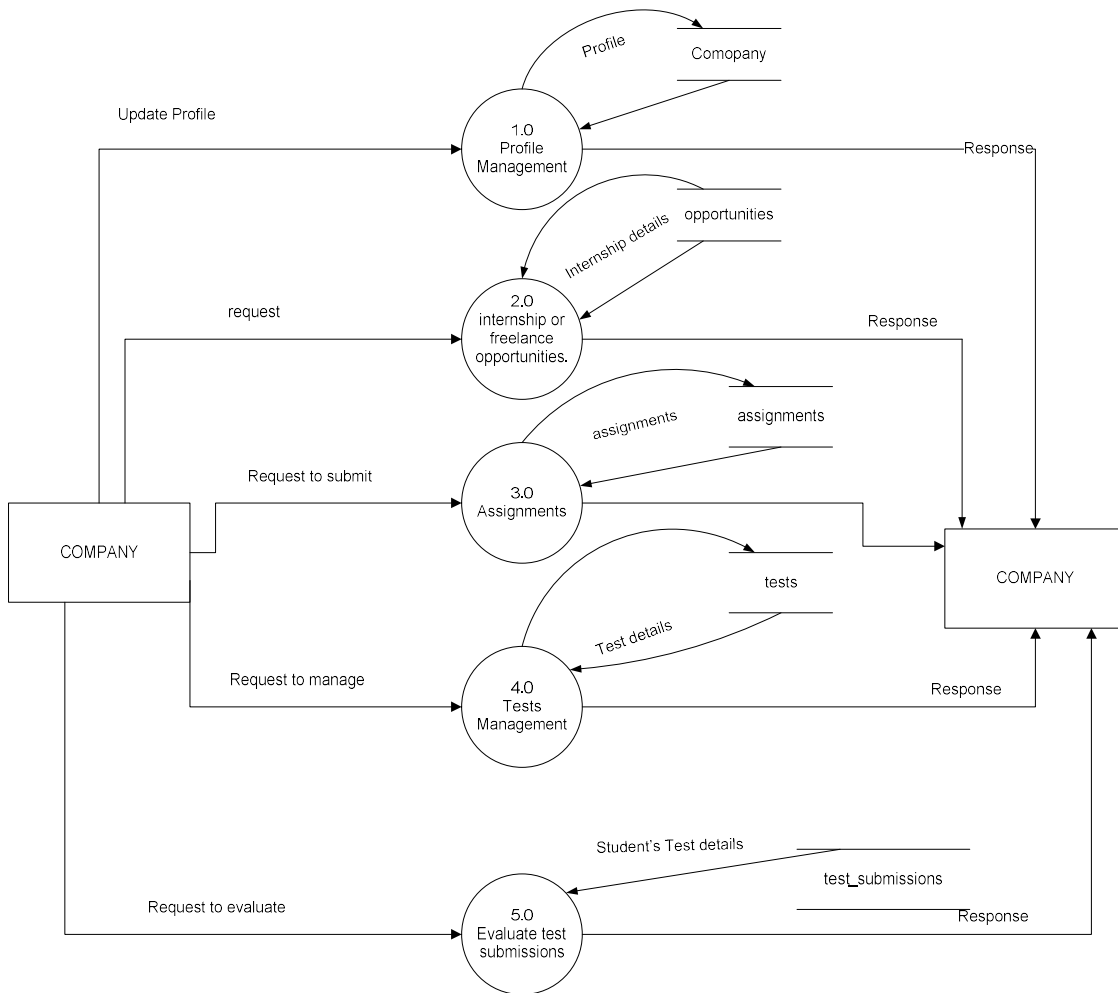
SECOND LEVEL DFD FOR ADMIN 1.0



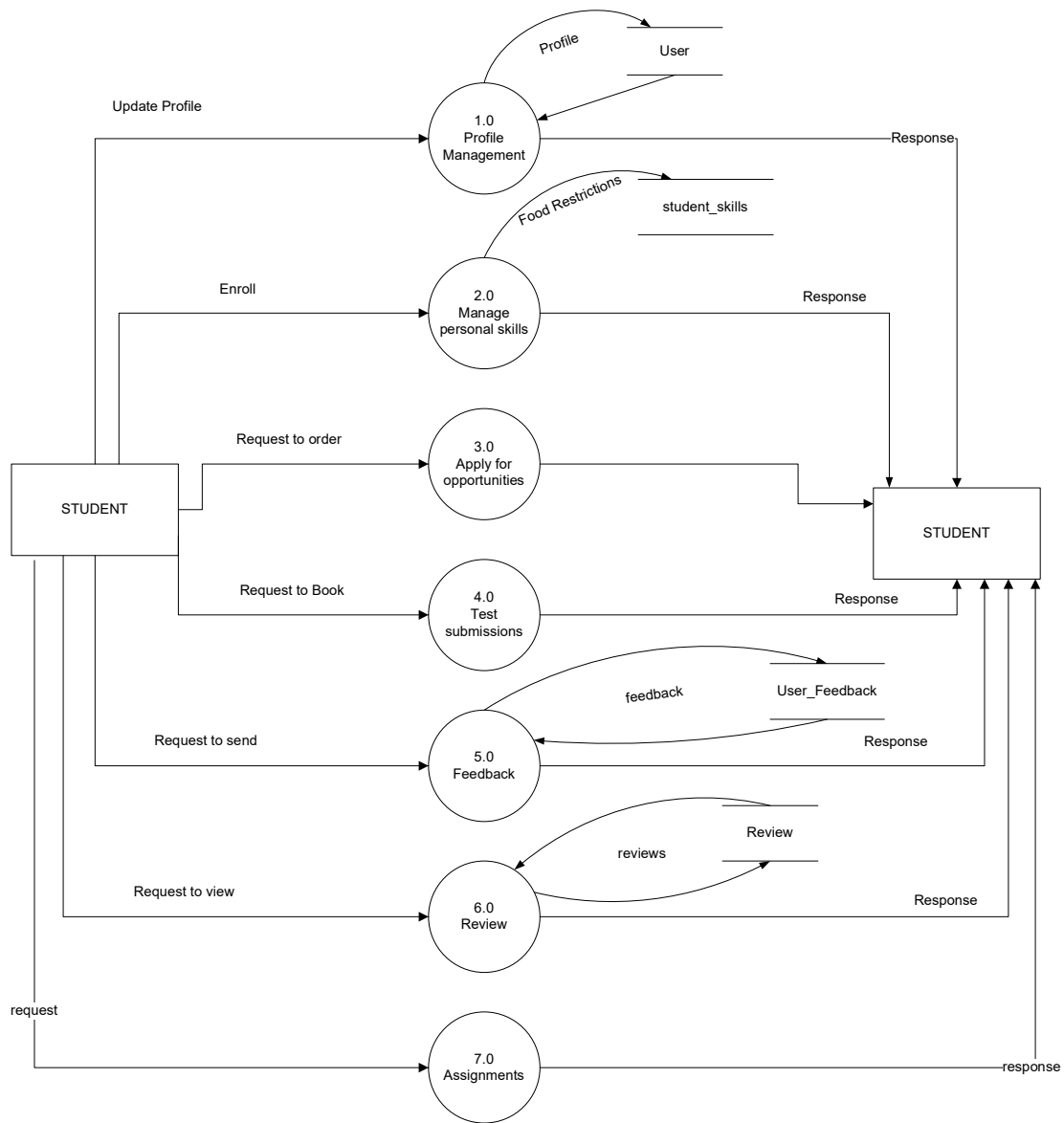
SECOND LEVEL DFD FOR ADMIN 5.0



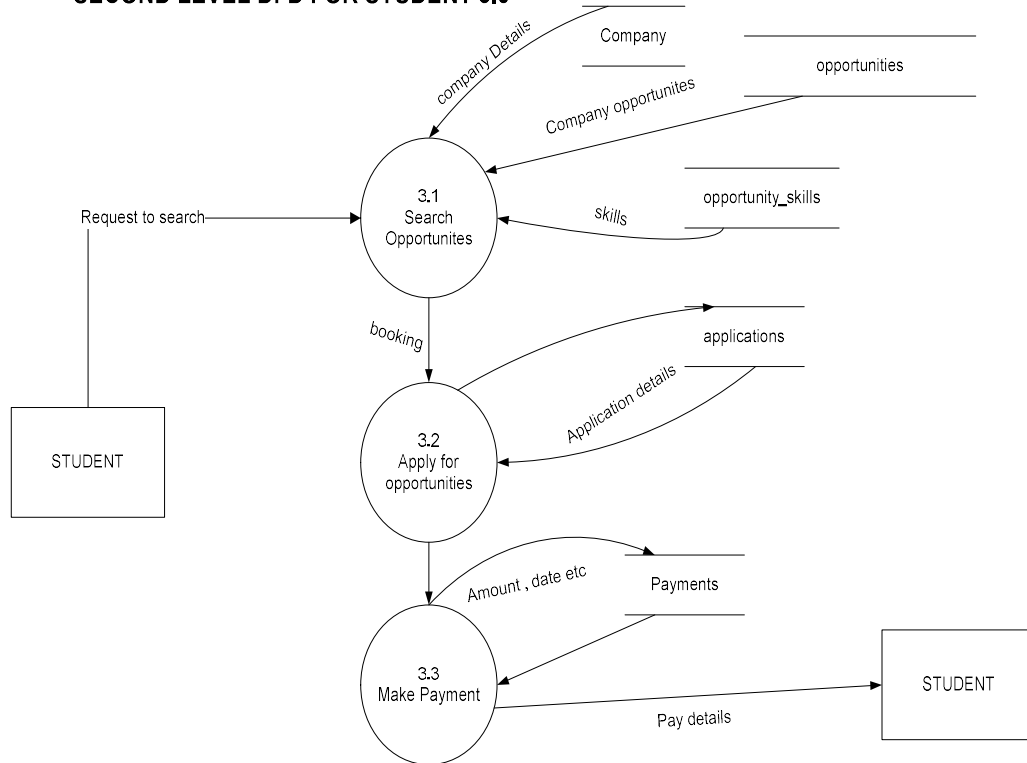
FIRST LEVEL DFD FOR COMPANY



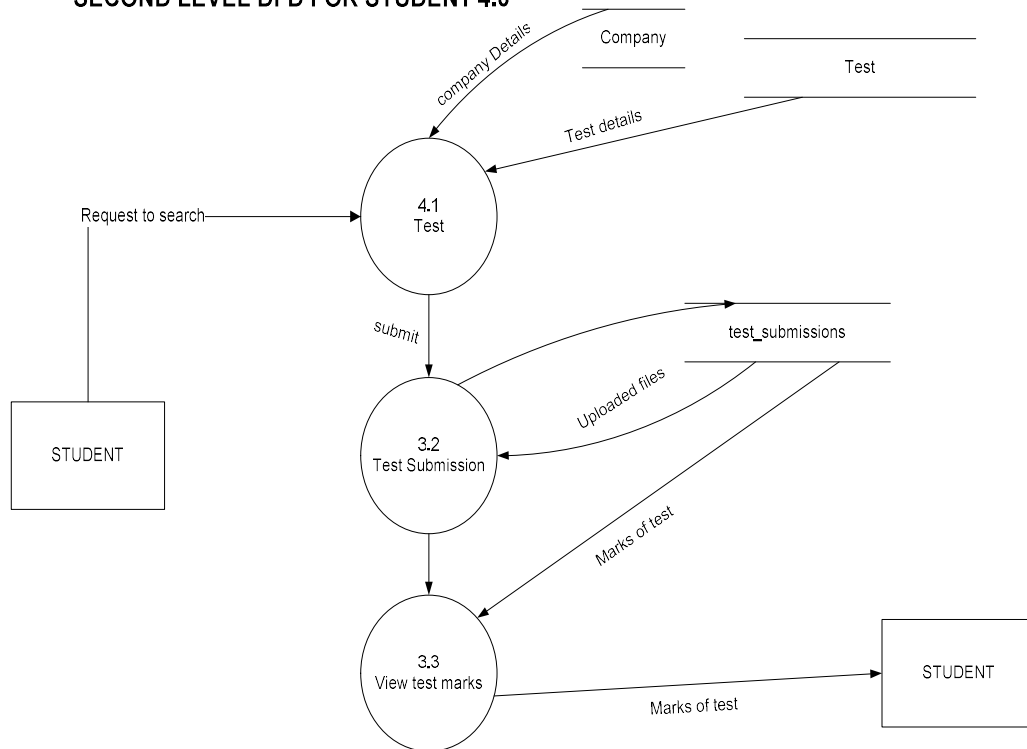
FIRST LEVEL DFD FOR STUDENT



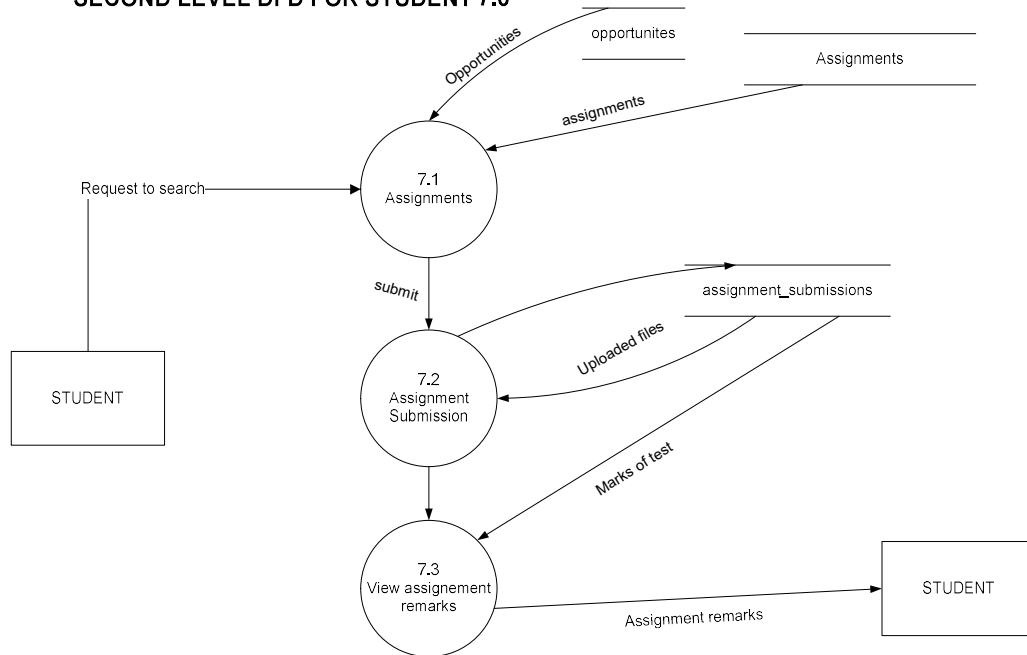
SECOND LEVEL DFD FOR STUDENT 3.0

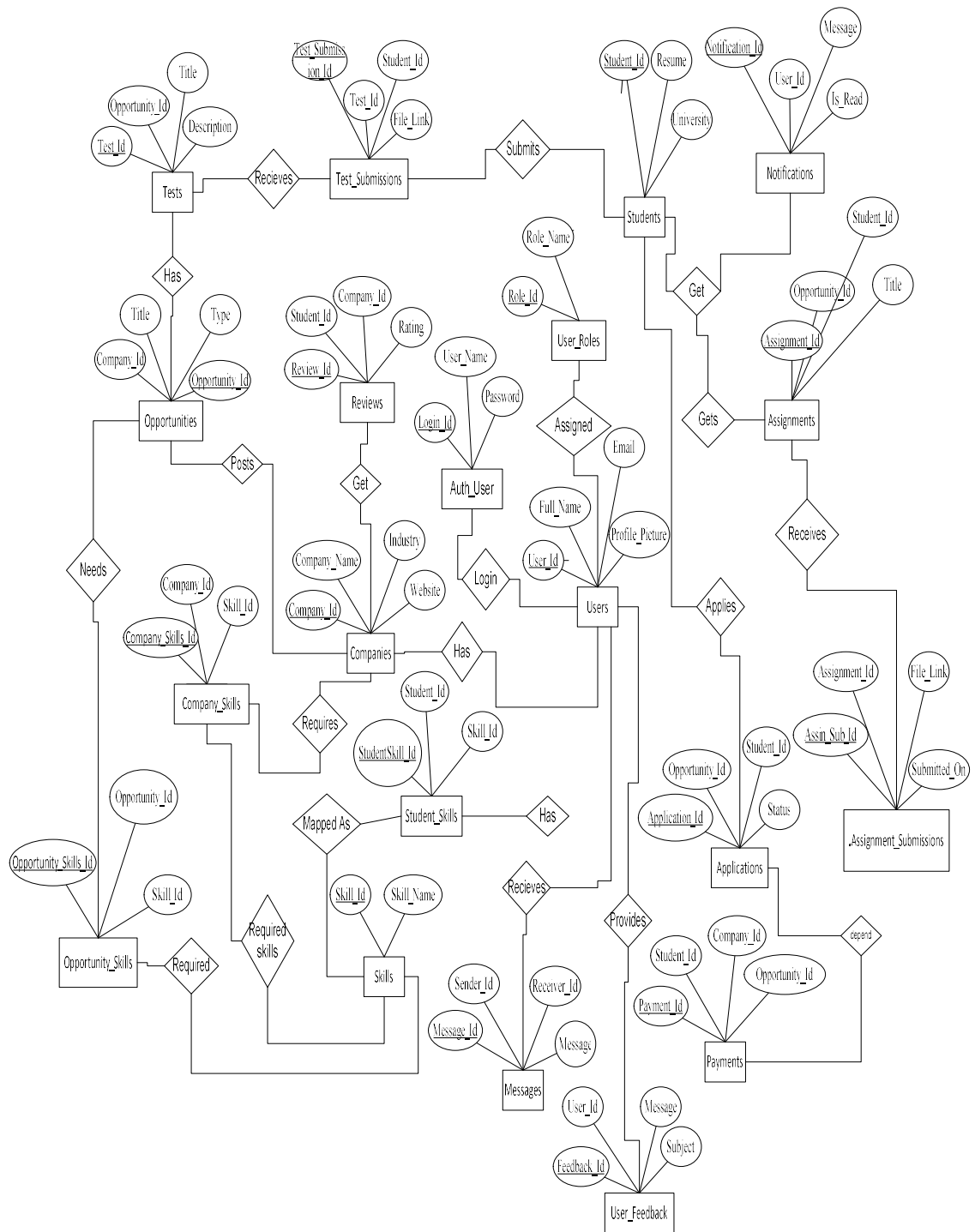


SECOND LEVEL DFD FOR STUDENT 4.0

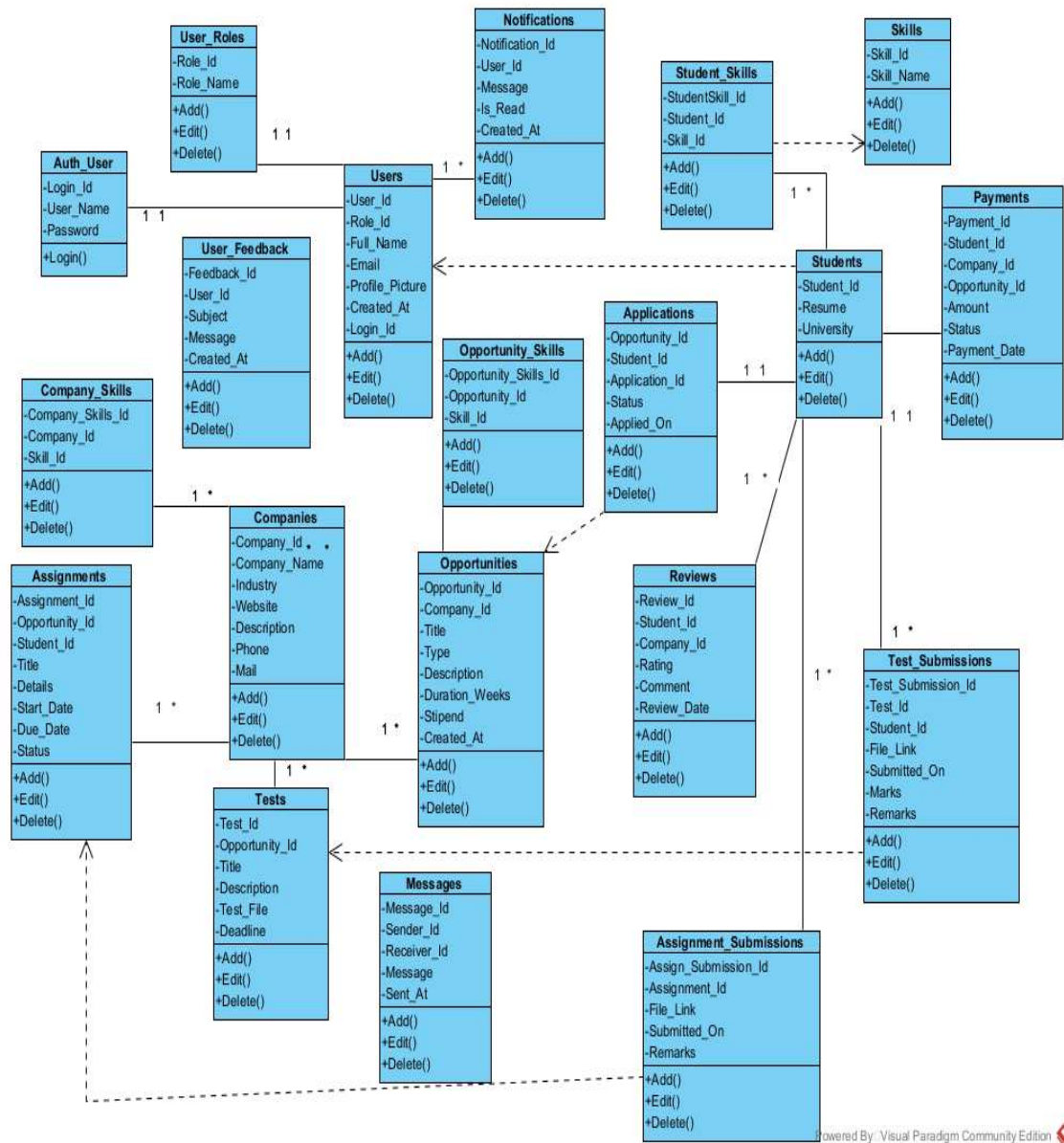


SECOND LEVEL DFD FOR STUDENT 7.0

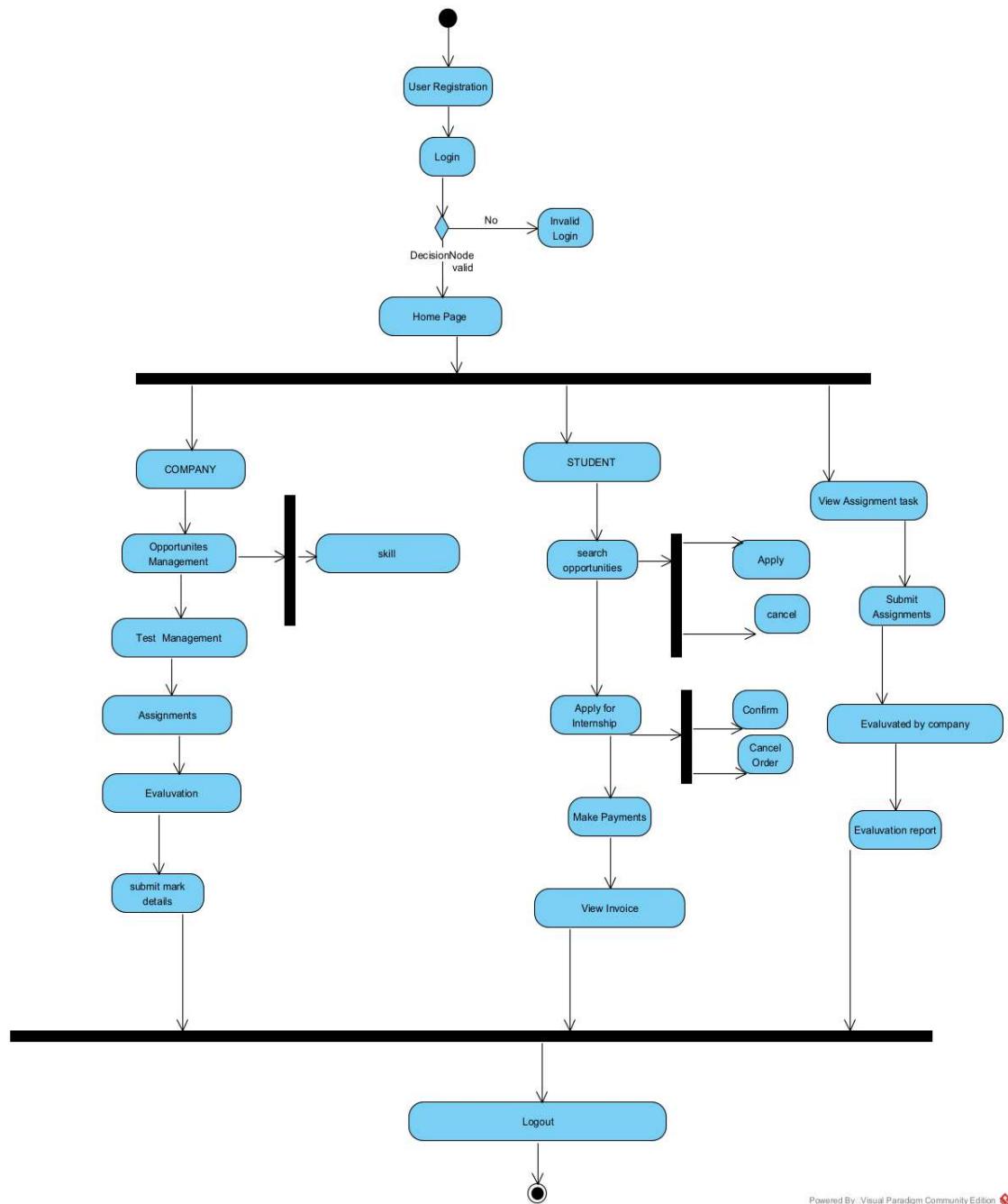




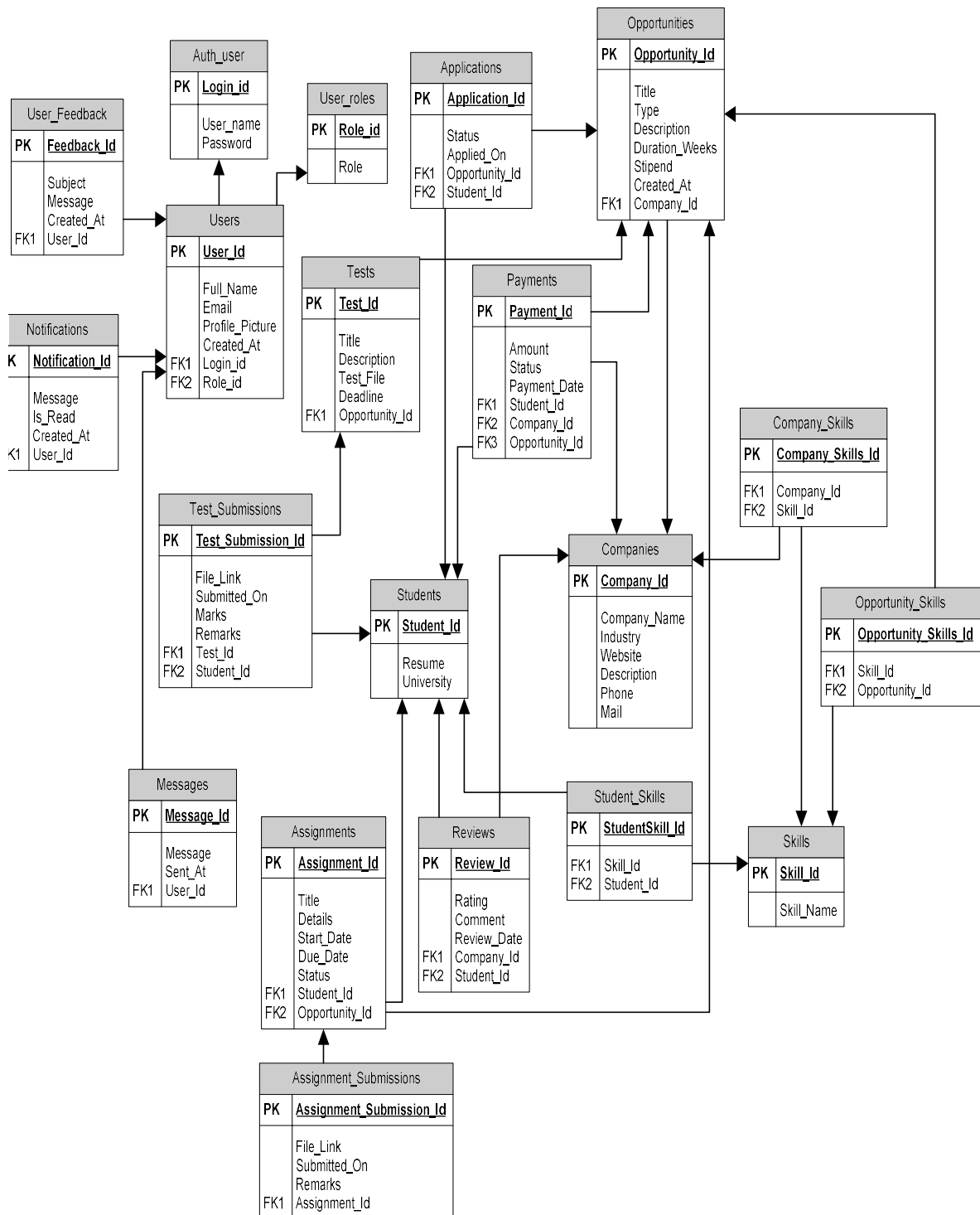
7.3 CLASS DIAGRAM



7.4 ACTIVITY DIAGRAM



8. COMPLETE DATABASE DESIGN



9. COMPLETE STRUCTURE OF THE PROJECT

9.1 MODULES AND THEIR DESCRIPTION

In the proposed **Online Internship & Freelance Portal**, the system functionalities are organized into three major roles: **Admin, Company, and Student**. Each role has specific responsibilities that collectively support internship postings, freelance project management, applications, tasks, communication, payments, and user engagement.

ADMIN MODULE

The Admin plays a central supervisory and management role in the Online Internship & Freelance Portal. Admin is responsible for managing all user accounts, including students and companies, by activating, deactivating, or reviewing profiles to ensure authenticity and system integrity. They handle role assignments, authentication verification, and monitor all login activities using the Auth_user module.

Admin oversees the entire ecosystem of internships and freelance opportunities. This involves reviewing opportunities posted by companies, removing inappropriate or expired listings, and ensuring that required skills match accepted standards. Admin also monitors student applications, test submissions, assignment progress, and hiring outcomes, helping maintain a high-quality recruitment workflow.

Additionally, the Admin is responsible for monitoring and verifying all payments within the platform—including student payments, company transactions, and opportunity-related fees. Admin manages the feedback and review system, ensuring transparency and trust between users. The reporting dashboard provides Admin with insights into overall system performance, including the number of users, active opportunities, total applications, successful hires, revenue tracking, and system engagement analytics. Notifications, user feedback management, and maintaining platform security also fall under the Admin's responsibility, ensuring smooth operation of the entire portal.

COMPANY MODULE

The Company module is designed for organizations seeking interns or freelancers. Companies manage their profile details, specialization, industry information, and skill requirements through an intuitive interface. They can maintain a list of company-specific skills, which helps in mapping relevant candidates and automating the filtering process.

Companies can create and post internship or freelance opportunities, specifying required skills, stipends, durations, test requirements, and project details. They can review applications submitted by students, shortlist candidates, reject profiles, or mark hiring decisions. The module also allows companies to schedule and manage skills-based tests by uploading files, setting deadlines, and evaluating submissions by providing marks and remarks.

For further evaluation, companies can assign real-time assignments or freelance tasks to selected students, track submissions, and update assignment statuses. A built-in messaging system allows seamless communication with students for interview scheduling, clarification, or project discussions. Companies can also view payment histories related to their

opportunities and track financial interactions made by students. Access to reviews and ratings left by students helps companies maintain transparency and improve their hiring practices. Overall, the Company module serves as a full-featured recruitment and project execution environment.

STUDENT MODULE

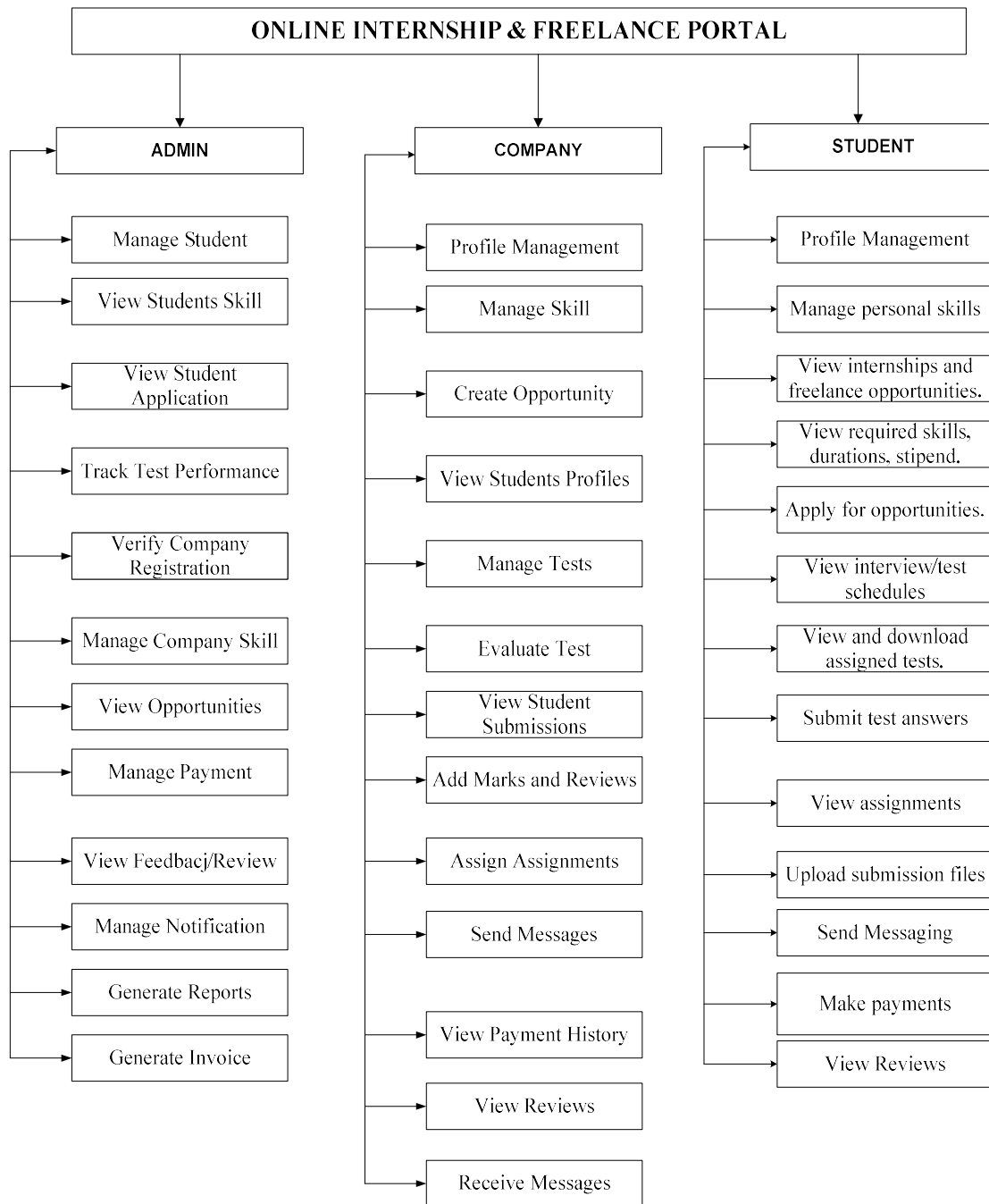
The Student module is designed to support learners and job-seekers by enabling them to manage their profiles, resumes, and skill sets. Students can update their personal skills, upload resumes, and enhance their visibility to companies looking for specific talent. The module provides the ability to browse and filter internship and freelance opportunities based on skills, duration, stipend, or job type.

Students can apply for opportunities directly through the system and track the status of their applications—from applied to shortlisted, hired, or rejected. For skill-based evaluation, students can access assigned tests, download test files, upload submissions, and later view marks and instructor remarks. Similarly, students can view project assignments given by companies and upload deliverables before the deadline.

A secure messaging system allows students to communicate directly with company representatives, enabling clarification about job details or project expectations. Students can also make payments when required, view payment receipts, and track their transaction history. They are encouraged to provide reviews and ratings for companies after their internship or freelance experience, contributing to platform transparency.

Additionally, students receive real-time notifications about application updates, test deadlines, assignment progress, chat messages, and payment status, ensuring they never miss important updates. The system also includes a feedback feature where students can submit suggestions or issues to the admin, supporting continuous improvement of the platform.

9.2 STRUCTURE OF MODULES



9.3 DATA STRUCTURE

1. Auth_User

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Login_Id	Int	4	Primary Key	Unique Login ID
User_Name	Varchar	50	Not Null	Username For Login
Password	Varchar	100	Not Null	Encrypted Password

2. User_Roles

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Role_Id	Int	4	Primary Key, Auto Increment	Unique Role ID
Role_Name	Varchar	50	Unique, Not Null	Student / Company / Admin

3. Users

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
User_Id	Int	4	Primary Key, Auto Increment	Unique User ID
Role_Id	Int	4	Foreign Key → User_Roles	Role Of The User
Full Name	Varchar	100	Not Null	Full Name Of User
Email	Varchar	150	Unique, Not Null	User Email
Profile Picture	Varchar	255	Null	Profile Image
Created_At	Timestamp	-	Default Current Timestamp	Account Creation Time
Login_Id	Int	4	Foreign Key → Auth_User	Login Credentials Link

4. Students

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Student_Id	Int	4	Primary Key, Foreign Key → Users.User_Id	Unique Student ID
Resume	Varchar	255	Null	Resume File
University	Varchar	150	Not Null	University Name

5. Companies

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Company_Id	Int	4	Primary Key, Foreign Key → Users.User_Id	Company User ID
Company_Name	Varchar	150	Not Null	Name Of Company
Industry	Varchar	100	Not Null	Industry Type
Website	Varchar	150	Null	Company Website
Description	Text	-	Not Null	Company Description
Phone	Bigint	10	Not Null	Contact Number
Mail	Varchar	150	Not Null	Company Email

6. Skills

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Skill_Id	Int	4	Primary Key, Auto Increment	Skill ID
Skill Name	Varchar	100	Unique, Not Null	Name Of Skill

7. Student_Skills

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
StudentSkill_Id	Int	4	Primary Key, Auto Increment	Skill Mapping ID
Student_Id	Int	4	Foreign Key → Students	Student
Skill_Id	Int	4	Foreign Key → Skills	Skill

8. Company_Skills

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Company_Skills_Id	Int	4	Primary Key, Auto Increment	Mapping ID
Company_Id	Int	4	Foreign Key → Companies	Company
Skill Id	Int	4	Foreign Key → Skills	Skill

9. Opportunities

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Opportunity_Id	Int	4	Primary Key, Auto Increment	Unique Opportunity ID
Company_Id	Int	4	Foreign Key → Companies	Posted By Company
Title	Varchar	100	Not Null	Opportunity Title
Type	Enum	-	Internship / Freelance	Type Of Opportunity
Description	Text	-	Not Null	Details Of Opportunity
Duration_Weeks	Int	4	Not Null	Duration In Weeks
Stipend	Decimal	(10,2)	Null	Stipend Amount
Created_At	Timestamp	-	Default Current Timestamp	Created Time

10. Opportunity_Skills

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
------------	----------	------	-------------	--------------

Opportunity_Skills_Id	Int	4	Primary Key, Auto Increment	Mapping ID
Opportunity_Id	Int	4	Foreign Key → Opportunities	Opportunity
Skill_Id	Int	4	Foreign Key → Skills	Skill Required

11. Applications

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Application_Id	Int	4	Primary Key, Auto Increment	Application ID
Opportunity_Id	Int	4	Foreign Key → Opportunities	Opportunity Applied
Student_Id	Int	4	Foreign Key → Students	Applicant
Status	Enum	-	Applied / Shortlisted / Hired / Rejected	Application Status
Applied_On	Date	-	Not Null	Application Date

12. Tests

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Test_Id	Int	4	Primary Key, Auto Increment	Test ID
Opportunity_Id	Int	4	Foreign Key → Opportunities	Related Opportunity
Title	Varchar	100	Not Null	Test Title
Description	Text	-	Not Null	Test Description
Test File	Varchar	255	Null	Test File
Deadline	Date	-	Not Null	Submission Deadline

13. Test_Submissions

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Test_Submission_Id	Int	4	Primary Key, Auto Increment	Submission ID
Test_Id	Int	4	Foreign Key → Tests	Test
Student_Id	Int	4	Foreign Key → Students	Student
File Link	Varchar	255	Not Null	Submission File
Submitted_On	Timestamp	-	Not Null	Submission Time
Marks	Decimal	(5,2)	Null	Marks Awarded
Remarks	Text	-	Null	Feedback

14. Assignments

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Assignment_Id	Int	4	Primary Key, Auto Increment	Assignment ID
Opportunity_Id	Int	4	Foreign Key → Opportunities	Related Opportunity
Student_Id	Int	4	Foreign Key → Students	Student Assigned
Title	Varchar	100	Not Null	Assignment Title
Details	Text	-	Not Null	Assignment Details
Start_Date	Date	-	Not Null	Start Date
Due_Date	Date	-	Not Null	Due Date
Status	Enum	-	Pending / Submitted / Completed	Assignment Status

15. Assignment_Submissions

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Assignment_Submission_Id	Int	4	Primary Key, Auto Increment	Submission ID
Assignment_Id	Int	4	Foreign Key → Assignments	Assignment
File_Link	Varchar	255	Not Null	Submission File
Submitted_On	Date	-	Not Null	Submission Date
Remarks	Text	-	Null	Feedback

16. Messages

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Message_Id	Int	4	Primary Key, Auto Increment	Message ID
Sender_Id	Int	4	Foreign Key → Users	Sender User
Receiver_Id	Int	4	Foreign Key → Users	Receiver User
Message	Text	-	Not Null	Message Content
Sent_At	Timestamp	-	Not Null	Sent Time

17. Payments

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Payment_Id	Int	4	Primary Key, Auto Increment	Payment ID
Student_Id	Int	4	Foreign Key → Students	Student
Company_Id	Int	4	Foreign Key → Companies	Company

Opportunity_Id	Int	4	Foreign Key → Opportunities	Related Opportunity
Amount	Decimal	(10,2)	Not Null	Payment Amount
Status	Enum	-	Pending / Paid / Failed	Payment Status
Payment_Date	Timestamp	-	Not Null	Payment Date

18. Reviews

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Review_Id	Int	4	Primary Key, Auto Increment	Review ID
Student_Id	Int	4	Foreign Key → Students	Reviewer
Company_Id	Int	4	Foreign Key → Companies	Company Reviewed
Rating	Int	1	Not Null	Rating (1–5)
Comment	Text	-	Null	Review Comment
Review_Date	Date	-	Not Null	Review Date

19. Notifications

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Notification_Id	Int	4	Primary Key, Auto Increment	Notification ID
User_Id	Int	4	Foreign Key → Users	User Receiving Notification
Message	Text	-	Not Null	Notification Message
Is_Read	Boolean	1	Default False	Read Status
Created_At	Timestamp	-	Not Null	Time Created

20. User_Feedback

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTIONS
Feedback_Id	Int	4	Primary Key, Auto Increment	Feedback ID
User_Id	Int	4	Foreign Key → Users	Feedback From User
Subject	Varchar	100	Not Null	Feedback Subject
Message	Text	-	Not Null	Feedback Message
Created_At	Timestamp	-	Not Null	Submitted Time

9.4 PROCESS LOGIC FOR EACH MODULE

1. Authentication & User Management

- **Input:** Username, Password, Role (Admin / Company / Student).

- **Process:**
Validate login credentials → Verify role from Auth_user table → Start secure session → Redirect to appropriate dashboard.
- **Output:** Authenticated access to Admin, Company, or Student dashboard with role-specific privileges.

2. Admin Module

- **Input:** Admin login details.
- **Process:**
 - Manage users (activate/deactivate students and companies).
 - Verify company profiles and approve new registrations.
 - View and manage all opportunities posted by companies.
 - Monitor student applications, test submissions, and assignment progress.
 - Oversee payments (pending, failed, completed).
 - Manage skills database, student skill sets, and company skill requirements.
 - Analyze system statistics through an admin dashboard (users, opportunities, applications, hires, payments).
 - Review feedback and take necessary actions.
 - Send system-wide notifications.
- **Output:** Streamlined platform with centralized admin control.

3. Company Module

- **Input:** Company login, opportunity details, test/assignment files.
- **Process:**
 - Update company profile and add company-specific skills.
 - Create and manage internship/freelance opportunities.
 - Define required skills for each opportunity.
 - Review student applications and shortlist/reject candidates.
 - Create tests for opportunities and upload related documents.
 - View and evaluate student test submissions by assigning marks and remarks.
 - Assign project tasks (assignments) and review student submissions.
 - Communicate with students through built-in messaging.
 - Monitor payments associated with selected opportunities.
 - View student reviews and feedback.
- **Output:** Well-managed recruitment cycle from posting to selection and task completion.

4. Student Module

- **Input:** Student registration, skills, resume, opportunity selection.
- **Process:**
 - Manage personal profile and update skill sets.
 - Browse internships and freelance opportunities.
 - Apply for opportunities based on interest and matching skills.
 - View application status (applied → shortlisted → hired/rejected).
 - Access and submit tests before deadlines.
 - View assignments and upload completed work.

- Communicate with companies through messaging.
- Make payments (if applicable) and view receipts.
- Provide reviews about companies after completion.
- Submit system feedback to the admin.
- Receive alerts on applications, tests, assignments, messages, and payments.
- **Output:** Successful submissions, application tracking, engagement with companies, and learning enhancement.

9.5 IMPLEMENTATION METHODOLOGY

The **Online Internship & Freelance Portal** is implemented using the **Agile Development Model**, enabling continuous improvements and frequent updates.

Agile Methodology Highlights:

- **Iterative & Incremental Development:** The project is divided into small, manageable modules such as authentication, opportunity posting, application handling, tests, assignments, and payments.
- **Continuous User Feedback:** Students and companies provide feedback that helps refine the recruitment workflow.
- **Flexibility:** Agile allows changes in requirements like adding new skills, updating opportunity structures, or enhancing messaging features.
- **Collaboration:** Admin, company, and student modules are developed and tested collaboratively through multiple sprints.

Agile ensures the portal evolves smoothly with high adaptability and user satisfaction.

9.6 LIST OF REPORTS

1. Admin Reports

- **User Enrollment Report:** Total number of registered students and companies with activation status.
- **Opportunity Report:** All internship and freelance opportunities posted by companies, with status (active/expired).
- **Application Report:** Number of applications, shortlisted candidates, and hired students.
- **Payment Report:** Payment transactions with date, status, student-company linkage, and total revenue.
- **Review & Feedback Report:** Summary of student reviews and reported system feedback.
- **Skill Distribution Report:** List of skills, number of students possessing them, and company skill requirements.

2. Company Reports

- **Application Status Report:** Summary of applications received per opportunity with student details.
- **Test Performance Report:** Student test submissions with marks and remarks.

- **Assignment Submission Report:** Assignment progress, submission dates, and completion status.
- **Opportunity Analytics Report:** Views, applications, hires, and performance of posted opportunities.
- **Payment History Report:** Payments made by students for internships or freelance tasks.

10. IMPLEMENTATION OF SECURITY MECHANISMS

In the Online Internship & Freelance Portal, strong security mechanisms are implemented to ensure data protection, user privacy, and safe transactions.

Key Security Measures Implemented:

1. Authentication & Password Security

- Passwords are encrypted

2. Role-Based Access Control

- Strict separation of roles: Admin, Company, Student.
- Unauthorized access is blocked using Django decorators and middleware.

3. Input & Data Protection

- CSRF tokens to prevent cross-site request forgery.
- Form validation to block malicious input, SQL injection, and XSS.

4. Secure Sessions

- Use of secure cookies (HttpOnly, SameSite, Secure).
- Session ID regeneration after login to prevent hijacking.

5. File Upload Security

- Allowed file type restrictions (resume, test submissions, assignments).
- Stored securely with randomized filenames.

6. Database Security

- Parameterized queries and ORM to avoid injection attacks.
- Role permissions implemented on database level.

These features ensure a reliable, safe, and secure recruitment environment for students and companies.

1. FUTURE SCOPE AND ENHANCEMENT OF THE PROJECT

The Online Internship & Freelance Portal can be expanded in many innovative ways to enhance user engagement, efficiency, and automation.

Future Enhancements:

- **AI-Based Recommendation System:** Personalized opportunity suggestions for students based on skills, history, and performance.
- **Automated Resume Screening:** AI to filter and shortlist applicants.
- **Chatbot for Support:** 24/7 virtual assistant to guide students and companies.
- **Mobile App Integration:** Android and iOS applications for better accessibility.
- **Video Interview Module:** Built-in video call feature for remote interviews.
- **Gamified Learning:** Skill quizzes, badges, and progress tracking.
- **Advanced Analytics:** Predict hiring trends, opportunity performance, and student skill gaps.
- **Blockchain Certification:** Verified certificate issuance for internships and freelance project completion.
- **Community Forums:** Company–student discussion rooms, Q&A spaces, peer help.
- **Real-Time Notification System:** Push notifications for deadlines, messages, and updates.

These upgrades will help transform the platform into a fully intelligent, future-ready digital recruitment ecosystem.

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